UTM
UNIVERSITI TEKNOLOGI MALAYSIA

COURSE INFORMATION

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School/Faculty:	**Faculty of Computing**		
Program name:	**Bachelor in Data Engineering with Honours**		
Course code:	**SCST1053**	**Academic Session/Semester:**	**2026 2027 1**
Course name:	**Introduction to Cloud Computing**	**Pre/co requisite (course name and code, if applicable):**	
Credit hours:	**3**		

Course synopsis	This foundational course introduces the principles, platforms, and practices of cloud computing, with an emphasis on practical applications and organisational requirements. Topics include cloud characteristics; service models, namely Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS); deployment models; cloud economics; security and the shared responsibility model; networking; computing; storage; managed databases; monitoring; scalability; and cloud architecture best practices. Students will explore the cloud service provider landscape and gain practical experience using Amazon Web Services (AWS), including deploying services, configuring automatic scaling, and monitoring cloud resources. Case studies will be used to examine how cloud architecture decisions affect cost, security, performance, reliability, and business requirements. The course will be delivered through lectures, hands-on laboratory sessions, active learning, industry talks, and project-based activities that promote collaboration and problem-solving. Upon successful completion of the course, students will be able to explain and compare fundamental cloud computing concepts, configure and deploy essential cloud services, and design basic scalable and secure cloud solutions that meet specified requirements. Students will also be able to evaluate fundamental cloud architecture decisions using industry-relevant tools and recognised best practices.
Course coordinator (if applicable)	Dr. Haslina Binti Hashim

Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail @utm.my**
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Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	1CLO	PLO 2(MQF Cluster Code)	3Taxonomies & 4Graduate attributes	5ESD, A & F	6VBE	T&L methods	Assessment methods
CLO1	Apply cloud computing concepts to solve computing problems using system thinking.	PLO1 (C1)	C3 (SC1)			Interactive, Scenario- and Case-Based Active Learning, Interactive Industry Talk, Experiential Industry Visit	Scenario-Based Quiz (AWS Module Knowledge Check) 10%, Final Examination (30%)
CLO2	Construct basic cloud computing services on a cloud platform using system thinking to ensure alignment with established best practices and standards.	PLO6 (C3D)	P3 A(IT2)			Lecture, Active Learning, Programming Lab, Interactive Industry Talk, Scaffolded AWS Cloud Laboratory	Individual AWS Practical Assignment (AWS Lab) (10%), Cloud Solution Project — Technical Deployment Component (10%), AWS Certification (5%), Final Examination (10%)
CLO3	Propose innovative cloud computing solutions collaboratively to inform strategic business solution.	PLO8 (C3F)	A3 A(CC2)			Scaffolded Project-Based Learning, Peer Learning, Presentation and Reflection	Cloud Solution Project — Solution Rationale and Teamwork Component (5%), Project Presentation (5%), Moderated Peer Assessment and Individual Reflection (5%), Visit Group Assignment Report and Presentation (10%)

This is the basic mapping required for the CI. Any added information is allowed (extra columns for weight or other elements) provided this is made consistent for all CI at program/school/faculty level.

1 Up to 5 CLO

2MQF Cluster Code

C1 = Knowledge & Understanding, C2 = Cognitive Skills, C3A = Practical Skills, C3B = Interpersonal Skills, C3C= Communication Skills, C3D = Digital Skills, C3E = Numercy Skills, C3F = Leadership, Autonomy & Responsibility, C4A = Personal Skills, C4B = Enterpreneurial Skills, C5 = Ethics & Professionalism

3Taxonomies of Learning Domain and 4UTM's Graduate Attributes for measurement of outcomes achievement

5Education for Sustainable Development (ESD), Flexibility (F) & Adaptability (A)

Systems Thinking (ST); Anticipatory Thinking (AT); Normative (N); Strategic Thinking (STT); Collaboration (C); Critical Thinking (CT); Self-Awareness (SA; Integrated Problem Solving (IPS)

6Value-Based Education (VBE)

Integrity; Synergy; Excellence; Sustainability

7 T – Test; Q – Quiz; HW – Homework; Asg – Assignment; PR – Project; Pr – Presentation; F – Final Exam etc. Please specify only the related type of assessment

Details on Innovative T&L practices:

No.	Type	Implementation

1	Interactive, Scenario- and Case-Based Active Learning	Students progress from short interactive lectures and worked examples to concept checks, Think–Pair–Share, Jigsaw, Gallery Walk and guided cloud cases. They compare cloud models and justify architecture, security, reliability and cost decisions with immediate lecturer and peer feedback.
2	Scaffolded Project-Based Learning	Teams address a bounded business scenario through staged milestones: requirements, simple cloud architecture, AWS prototype, technical validation, security and cost rationale, and final presentation. Feedback is provided at each checkpoint, with defined team roles and individually attributable contributions.
3	Scaffolded AWS Cloud Laboratory	Laboratory activities follow a demonstrate–practise–apply sequence: lecturer demonstration, guided configuration and an individual challenge. Students configure, deploy, validate and troubleshoot basic compute, storage, networking, database, security and monitoring services, supported by lab check-offs and formative feedback.
4	Experiential Industry Visit	Students receive a pre-visit briefing and guided observation questions before visiting a data centre, cloud operations facility or suitable virtual alternative. They engage with industry professionals and complete a post-visit architecture map or reflection connecting their observations to course concepts and best practices.
5	Programming Lab	Students will perform lab exercise via selected cloud platform.
6	Interactive Industry Talk	Students review introductory material and prepare questions before an industry professional presents a real cloud project, architectural decisions, operational challenges and career pathways. Students subsequently analyse the case in small groups and record the key lessons applicable to their project or laboratory work.

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :

Apply systematic problem-solving and digital skills to select, configure and troubleshoot appropriate cloud services for a defined problem.

Collaborate responsibly in a team by communicating ideas, coordinating tasks, sharing knowledge and providing constructive peer guidance and feedback.

Present and justify technical decisions clearly using written, visual and oral communication, and use feedback to improve individual and team performance.

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO* Compulsory	Learning and Teaching Activities									TOTAL SLT
			Face-to-Face (F2F)							Non F2F Independent Learning		
			Physical				Online (Synchronous)			Online (Asynchronous)	Others	
			L	T	P	O	L	T	P	O		
Week 1	**1.0 Introduction to Cloud Computing** - Definition of cloud computing - Deployment Models: Public cloud, private cloud, hybrid cloud, community cloud - Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS) - Use cases for each service model	CLO1					3				2	2	7
Week 2	**2.0 Cloud Service Providers** - Key services and offerings of each provider - Strengths and unique features of each platform - Comparison of services (e.g., compute, storage, networking)	CLO1					3				2	2	7
Week 3	**3.0 Cloud Economics and Billing** - Fundamentals of Pricing - Total Cost of Ownership - Cloud Organizations - Cloud Billing - Cloud Technical Support Models	CLO2	2		1						3	2	8
Week 4	**4.0 Cloud Security** - Shared Responsibility Model - Identity and Access Management - Securing Accounts and Data - Compliance	CLO2	2		1						3	2	8
Week 5	**5.0 Cloud Networking and Content Delivery** - Networking basics - Virtual Private Cloud (VPC) - VPC Networking - VPC Security - VPC Domain Name System (DNS) - VPC Content Delivery Network (CDN)	CLO1	2		1						3	2	8
Week 6	6.0 Cloud Compute Services - Compute Services Overview - Elastic Computing (EC) - EC Cost Optimization - Container Services	CLO1	2		1						2	2	7
Week 7	- Serverless Compute Service - Deployment and Management of web application and services Industry Talk	CLO1			1		2				2	2	7
Week 8	MID SEMESTER BREAK												

Week 9	**7.0 Cloud Storage** - Cloud storage models: Object storage, block storage, file storage - Cloud storage services (AWS S3, Google Cloud Storage, Azure Blob Storage)	CLO1	2		1					2	2	7
Week 10	**Industry Visit**	CLO2	3							2	2	7
Week 11	**8.0 Cloud Databases** - Relational Database Services - NoSQL Database Services - Warehouse service	CLO2	2		1					2	2	7
Week 12	**8.0 Cloud Databases** - Relational Database Services - NoSQL Database Services - Warehouse service Industry Talk	CLO2			1		2			2	2	7
Week 13	**10.0 Auto Scaling and Monitoring** - Elastic Load Balancing - Comprehensive monitoring and observability service	CLO2	2		1					2	2	7
Week 14	**11.0 Case Studies and Industry Applications** - Real-world applications of cloud-based data warehousing	CLO3			1		2			2	3	8
Week 15		CLO3	1		2					2	3	8
										SUB-TOTAL SLT :		**103**

Continuous Assessment		**%**	**Face-to-Face (F2F)**		**NF2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online (Synchronous)**	**Online (Asynchronous)**	**Others**	
1	Scenario-Based Quiz (AWS Module Knowledge Check)	10	1				1
2	Individual AWS Practical Assignment (AWS Lab)	10			12	3	15
3	Visit Group Assignment Report and Presentation	10	2				2
4	Project Presentation	5	1				1
5	Technical Deployment Component	10			4	8	12
6	Solution Rationale and Teamwork Component	5	1		2		3
7	Moderated Peer Assessment and Individual Reflection	5			1		1
8	AWS Certification	5		1	5		6
Assessment activities carried out during the semester to monitor student progress (e.g., quizzes, assignments, projects).					**SUB-TOTAL SLT :**		**41**

Final Assessment		**%**	**Face-to-Face (F2F)**		**NF2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online (Synchronous)**	**Online (Asynchronous)**	**Others**	
1	Final Examination	40	3			3	6
2							0
3							0
4							0
5							0
Assessment conducted at the end of the course to evaluate the achievement of overall learning outcomes (e.g., final exam, final project).					**SUB-TOTAL SLT :**		**6**
					SLT for Assessment:		**47**
					GRAND TOTAL SLT:		**150**

A	% SLT for F2F Physical Component	26.39
B	% SLT for Online & Independent Learning Component :	73.61
C	%SLT for Online Component:	41.33
D	% SLT for All Practical Component:	8.33
D1	% SLT for F2F Physical Practical Component:	8.33
D2		0.00
Please tick (/) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)		

Identify special requirement or resources to deliver the course (e.g.,software,nursery, computer lab, simulation room etc)

Technology-enabled computer laboratory or classroom with one internet-connected computer or laptop per student, a current web browser, stable broadband Internet and projection/audio facilities. Institutional access to AWS Academy Cloud Foundations, individual learner accounts, the AWS Academy sandbox environment, AWS Management Console and UTM LMS are required for hands-on laboratories, demonstrations, videos and knowledge checks. The lecturer requires AWS Academy educator access for course materials, assessments and lab administration. Standard productivity, PDF-viewing and presentation software should be available. A text editor, terminal/SSH client or AWS CLI may be provided when required by a specific laboratory or project. Personal AWS accounts or credit cards are not required. Approved industry-partner access and appropriate visit arrangements are required where applicable.

References (include required and further readings, and should be the most current)

References
1. Comer, D. (2022). The cloud computing book: The future of computing explained. Chapman and Hall/CRC.
2. Zburivsky, D., & Partner, L. (2021). Designing cloud data platforms. Simon & Schuster.
3. Marinescu, D. C. (2022). Cloud computing: Theory and practice. Morgan Kaufmann.
4. Manvi, S., & Shyam, G. (2021). Cloud computing: Concepts and technologies. CRC Press.

Other additional information (if applicable)

Academic honesty and plagiarism: ***(Below is just a sample)***

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES)
Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (if applicable)

Disclaimer:

All teaching and learning materials associated with this course are for personal use only. The materials are intended for educational purposes only. Reproduction of the materials in any form for any purposes other than what it is intended for is prohibited.
While every effort has been made to ensure the accuracy of the information supplied herein, Universiti Teknologi Malaysia cannot be held responsible for any errors or omissions.

ELT = (Theory + Industrial Guidance + Assessment) x 50%

Total of credit for LI/Practical = ELT/40 Notional Hours
Note: For ODL Programme : Courses with mandatory practical requirement imposed by programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

Prepared by:	**Certified by:**
Name:	**Name:**
Signature:	**Signature:**
Date: 30-Aug-26	**Date:** 30-Aug-26